

$$\begin{array}{c}
\mathcal{K}_{0^+}^{\#1} \quad \mathcal{K}_{0^+}^{\#2} \\
\mathcal{K}_{0^+}^{\#1} \dagger \begin{array}{|c|c|} \hline k^2 \frac{(4)}{\kappa_1} & k^2 \frac{(4)}{\kappa_1} \\ \hline \end{array} \\
\mathcal{K}_{0^+}^{\#2} \dagger \begin{array}{|c|c|} \hline k^2 \frac{(4)}{\kappa_1} & k^2 \frac{(4)}{\kappa_1} \\ \hline \end{array} \quad \mathcal{K}_{1^+}^{\#1}{}_\alpha \quad \mathcal{K}_{1^+}^{\#2}{}_\alpha \\
\mathcal{K}_{1^+}^{\#1} \dagger^\alpha \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#2} \dagger^\alpha \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \quad \mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta} \\
\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta} \begin{array}{|c|} \hline 0 \\ \hline \end{array} \quad \mathcal{K}_{3^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{K}_{3^+}^{\#1} \dagger^{\alpha\beta\chi} \begin{array}{|c|} \hline 0 \\ \hline \end{array}
\end{array}$$

$$\begin{array}{c}
\mathcal{J}_{0^+}^{\#1} \quad \mathcal{J}_{0^+}^{\#2} \\
\mathcal{J}_{0^+}^{\#1} \dagger \begin{array}{|c|c|} \hline \frac{1}{4k^2 \frac{(4)}{\kappa_1}} & \frac{1}{4k^2 \frac{(4)}{\kappa_1}} \\ \hline \end{array} \\
\mathcal{J}_{0^+}^{\#2} \dagger \begin{array}{|c|c|} \hline \frac{1}{4k^2 \frac{(4)}{\kappa_1}} & \frac{1}{4k^2 \frac{(4)}{\kappa_1}} \\ \hline \end{array} \quad \mathcal{J}_{1^+}^{\#1}{}_\alpha \quad \mathcal{J}_{1^+}^{\#2}{}_\alpha \\
\mathcal{J}_{1^+}^{\#1} \dagger^\alpha \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#2} \dagger^\alpha \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \quad \mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta} \\
\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta} \begin{array}{|c|} \hline 0 \\ \hline \end{array} \quad \mathcal{J}_{3^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{J}_{3^+}^{\#1} \dagger^{\alpha\beta\chi} \begin{array}{|c|} \hline 0 \\ \hline \end{array}
\end{array}$$

Lagrangian

$\frac{(4)}{\kappa_1} \partial_\beta \mathcal{K}_{\chi}{}^\delta \partial^\chi \mathcal{K}^\alpha{}_\beta$		
Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}_{0^+}^{\#2} == 3 \mathcal{J}_{0^+}^{\#1}$	1	$2 \partial_\chi \partial_\beta \partial_\alpha \mathcal{J}^{\alpha\beta\chi} == \partial_\chi \partial^\chi \partial_\beta \mathcal{J}^\alpha{}_\alpha{}^\beta$
$\mathcal{J}_{1^+}^{\#2\alpha} == 0$	3	$\partial_\delta \partial_\chi \partial_\beta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta \partial^\delta \partial_\chi \partial^\alpha \mathcal{J}^{\alpha\beta}{}_\beta == \partial_\delta \partial^\delta \partial_\chi \partial^\alpha \mathcal{J}^{\beta}{}_\beta{}^\chi + \partial_\delta \partial^\delta \partial_\chi \partial_\beta \mathcal{J}^{\alpha\beta\chi}$
$\mathcal{J}_{1^+}^{\#1\alpha} == 0$	3	$\partial_\delta \partial_\chi \partial_\beta \partial^\alpha \mathcal{J}^{\beta\chi\delta} == \partial_\delta \partial^\delta \partial_\chi \partial_\beta \mathcal{J}^{\alpha\beta\chi}$
$\mathcal{J}_{2^+}^{\#1\alpha\beta} == 0$	5	$k^\chi (\partial_\epsilon \partial_\chi \partial^\beta \partial^\alpha \mathcal{J}^\delta{}_\delta{}^\epsilon + 2 \partial_\epsilon \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}^\delta{}_\chi{}^\epsilon - 3 \partial_\epsilon \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\beta\delta\epsilon} -$ $3 \partial_\epsilon \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\alpha\delta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial_\delta \partial_\chi \mathcal{J}^{\alpha\beta\delta} + \eta^{\alpha\beta} \partial_\phi \partial_\epsilon \partial_\delta \partial_\chi \mathcal{J}^{\delta\epsilon\phi} - \eta^{\alpha\beta} \partial_\phi \partial^\phi \partial_\epsilon \partial_\chi \mathcal{J}^\delta{}_\delta{}^\epsilon) == 0$
$\mathcal{J}_{3^+}^{\#1\alpha\beta\chi} == 0$	7	$k^\delta k^\epsilon (3 \partial_\epsilon \partial^\chi \partial^\beta \partial^\alpha \mathcal{J}^\phi{}_\phi{}^\delta - \partial_\epsilon \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}^{\chi\phi}{}_\phi - \partial_\epsilon \partial_\delta \partial^\chi \partial^\alpha \mathcal{J}^{\beta\phi}{}_\phi - \partial_\epsilon \partial_\delta \partial^\chi \partial^\beta \mathcal{J}^{\alpha\phi}{}_\phi + 2 \partial_\phi \partial^\chi \partial^\beta \partial^\alpha \mathcal{J}^{\delta\epsilon}{}_\phi -$ $4 \partial_\phi \partial_\epsilon \partial^\beta \partial^\alpha \mathcal{J}^\chi{}_\phi{}^\delta - 4 \partial_\phi \partial_\epsilon \partial^\chi \partial^\alpha \mathcal{J}^\beta{}_\phi{}^\delta - 4 \partial_\phi \partial_\epsilon \partial^\chi \partial^\beta \mathcal{J}^\alpha{}_\phi{}^\delta + 5 \partial_\phi \partial_\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\phi} + 5 \partial_\phi \partial_\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\phi} +$ $5 \partial_\phi \partial_\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\phi} - 5 \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\alpha\beta\chi} - \eta^{\beta\chi} \partial_\gamma \partial_\epsilon \partial_\delta \partial^\alpha \mathcal{J}^\phi{}_\phi{}^\gamma - \eta^{\alpha\chi} \partial_\gamma \partial_\epsilon \partial_\delta \partial^\beta \mathcal{J}^\phi{}_\phi{}^\gamma - \eta^{\alpha\beta} \partial_\gamma \partial_\epsilon \partial_\delta \partial^\chi \mathcal{J}^\phi{}_\phi{}^\gamma +$ $\eta^{\beta\chi} \partial_\gamma \partial_\phi \partial_\epsilon \partial^\alpha \mathcal{J}^\phi{}_\phi{}^\gamma + \eta^{\alpha\chi} \partial_\gamma \partial_\phi \partial_\epsilon \partial^\beta \mathcal{J}^\phi{}_\phi{}^\gamma + \eta^{\alpha\beta} \partial_\gamma \partial_\phi \partial_\epsilon \partial^\chi \mathcal{J}^\phi{}_\phi{}^\gamma - \eta^{\beta\chi} \partial_\gamma \partial_\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\alpha\phi\gamma} - \eta^{\alpha\chi} \partial_\gamma \partial_\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\beta\phi\gamma} -$ $\eta^{\alpha\beta} \partial_\gamma \partial_\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\chi\phi\gamma} + \eta^{\beta\chi} \partial_\gamma \partial^\gamma \partial_\epsilon \partial_\delta \mathcal{J}^{\alpha\phi}{}_\phi + \eta^{\alpha\chi} \partial_\gamma \partial^\gamma \partial_\epsilon \partial_\delta \mathcal{J}^{\beta\phi}{}_\phi + \eta^{\alpha\beta} \partial_\gamma \partial^\gamma \partial_\epsilon \partial_\delta \mathcal{J}^{\chi\phi}{}_\phi) == 0$
Total # constraint(s):	19	
Resolved unitarity condition(s):	True	